

3	Sessionals(May include Assignments/Projects/Tutorials/Quiz/ Lab evaluations)	30
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ELECTIVE COURSE-4.2

UME625: ALTERNATIVE FUELS FOR VEHICLES				
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	3	0	0	3
Course Objectives: To equip students with critical knowledge of environment friendly vehicle drive systems across various mobility applications. The course emphasizes sustainable energy sources, eco-friendly production, and high-performance powertrain optimization. Students will learn to analyse and compare advanced technologies for ultra-low emissions in Internal Combustion Engines (ICE) and hybrid systems, including the use of hydrogen fuel.				
Basis of Internal Combustion Engines: Important qualities of SI and CI engine fuels, Combustion in SI engines, Different phases of combustion, Thermodynamic analysis of SI & CI engine combustion, combustion efficiency and various losses, Basics of various systems of IC engine combustion system (intake, TC, injection, ignition, combustion, exhaust, EGR, lube, cooling), overview on problems for high efficiency & ultra-low emissions ICE, legislation, and possible actions, CO₂ & other emission reduction in IC engines. IC Engine performance basics. Alternate Fuels for Transportation Industry: Need, opportunity & Challenges: Need of alternative gaseous fuels, future automotive gaseous fuels, hydrogen, CNG, LNG, LPG. Stoichiometric air-fuel ratio, Physical properties of different gaseous fuels, mode of engine operations, spark ignition and dual fuel mode, multi fuel mode, combustion and performance of engines with air-fuel ratio, specific problems, safety and environmental aspects, economic aspects, production. CO₂ emissions overall balance. Well-to-wheel analysis. Decade of transition, Moving from oil based to gas based economy. Basics of EV & Hybrid Powertrain System: History of electric vehicles, impact of modern drive-trains on energy supplies; Configurations of Electric Vehicles; Concept of Hybrid Electric Drive Trains; Architectures of Hybrid Electric Drive Trains; Series Hybrid Electric Drive Trains; Parallel Hybrid Electric Drive Trains; Torque-Coupling and Speed-Coupling of Drive Trains.				

Approved in 109th meeting of the Senate held on March 16, 2023. Revised in 112th and 113th meeting of the Senate held on March 11 and September 7, 2024, respectively. Further, revision Approved in 1st meeting of Academic Council held on September 11, 2025.

Fuel cell application to propulsion systems: General overview on the electrochemical conversion process, fuel cell types and characteristics. Fuel cell application to powertrain systems: types, operating problems, performance, hydrogen generation and storage systems, energy and emissive balance; applications, technical and economic issues, further developments. Different ways to store the H₂: Physical and Chemical, Energy intensity for each storing type.

Virtual development techniques: Application of modelling techniques and parameterization processes for all parts of the propulsion system, involving software, 0D,1D,3D modelling, Problem definition and planning, tool selection based on problem statement, After treatment system. Case Study: The environmental performance of advanced propulsions systems for various regulations.

Course Learning Outcomes (CLOs)

After completion of this course, students will be able to:

1. Compare technical solutions for emissions control and CO₂ reduction in propulsion systems with a quantitative approach;
2. Apply theoretical basis and operating principles to enhance performance and minimize environmental impact of propulsion systems;
3. Identify feasible options for use of alternative fuels and energy vectors
4. Evaluate technical pathways using modelling techniques for the development of advanced propulsion systems.

Text Books

1. The Internal combustion Engine in theory and practice: C F Taylor, MIT Press, Cambridge.
2. Hydrogen Fuel for Surface Transportation: Joseph Norbeck, SAE Publications, 1996
3. Fuel cell fundamentals: O'hayre, Ryan, Suk-Won Cha, Whitney Colella, and Fritz B. Prinz, John Wiley & Sons, 2016.

Reference Books

1. Internal Combustion engine fundamentals: J B Heywood, Mc-Graw Hill Publications.
2. Fundamentals of internal combustion engines: Gill, Smith and Ziurys, Oxford and IBH.

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